PoopVolt / prototype electrical wiring

As built in PR #49 — 10 October 2026 — UNO Q on USB power, 3.3 V Grove ports, solar panel charges a LiPo through the Waveshare manager

Grove conductors: 1 yellow / 2 white / 3 red (VCC) / 4 black (GND)

P1
Laptop USB-C supply
5 V over USB-C
Data omitted
5_V
GND
VBUS
GND
+5V_USB
GND
USB_VBUS
USB_GND
USB_5_V
GND
U1
UNO Q + Base Shield
3.3_V_rail
3V3
+3V3
PB10_/_Wire_SCL
SCL
SCL
PB11_/_Wire_SDA
SDA
SDA
PB7_/_Serial1_RX
D0
D0_RX
PB6_/_Serial1_TX
D1
D1_TX
PA4_/_light_ADC
A0
LIGHT_A0
PA5_/_PV_current_ADC
A1
PV_ISENSE_A1
PA6_/_spare
A2
SPARE_A2
Base Shield VCC switch = 3.3 V
STM32U585 MCU / 3.3 V I/O
Wire = MCU I2C2
Serial1 = MCU USART1
Linux Bridge is internal UART
Bridge does not use D0 / D1
BASE SHIELD PORTS
J1
I2C port / AHT20
SCL 1 — SCL — 1 SCL
SDA 2 — SDA — 2 SDA
VCC 3 — +3V3 — 3 VCC
GND 4 — GND — 4 GND
CONNECTED GROVE MODULES
U2
Grove AHT20 / 0x38
Temperature
and humidity
J2
I2C port / LCD
SCL 1 — SCL — 1 SCL
SDA 2 — SDA — 2 SDA
VCC 3 — +3V3 — 3 VCC
GND 4 — GND — 4 GND
U3
Grove LCD 16x2 / 0x3E
White on Blue
No RGB controller
J3
UART port / PN532
D0_/_RX 1 — D0_RX — 1 TX
D1_/_TX 2 — D1_TX — 2 RX
VCC 3 — +3V3 — 3 VCC
GND 4 — GND — 4 GND
U4
Grove PN532 / HSU
HSU 115200 baud
Antenna attached
Phone reads NDEF
J4
A0 port / light
A0 1 — LIGHT_A0 — 1 OUT
A1 2 — PV_ISENSE_A1 — 2 NC_/_unused
VCC 3 — +3V3 — 3 VCC
GND 4 — GND — 4 GND
U5
Grove Light Sensor (P)
Pin 2: unused
A1 carries PV current
J5
A1 port / PV current
A1 1 — PV_ISENSE_A1 — 1 OUT
A2 2 — SPARE_A2 — 2 NC_/_unused
VCC 3 — +3V3 — 3 VCC
GND 4 — GND — 4 GND
U6
Grove 2.5A DC current (ACS70331)
800 mV/A
zero 266.5 mV
IP+ — PV_POS
IP− — PV_TO_CHARGER
SOLAR HARVESTING (separate from the UNO Q supply)
PV1
Mini solar panel R1123
90 x 60 mm
rating unverified
+ POS — PV_POS
− NEG — PV_NEG
U7
Waveshare Solar Power Manager (5 V / 1 A)
PV_TO_CHARGER — SOL+ — SOLAR_IN+
PV_NEG — SOL− — SOLAR_IN−
BAT_POS — BAT+ — BAT+(PH2.0)
BAT_NEG — BAT− — BAT−(PH2.0)
CN3791 MPPT charger
MPPT SET = 6 V
Battery switch ON
USB OUT not connected
BT1
LiPo 3.7 V 1000 mAh
+ POS — BAT_POS
− NEG — BAT_NEG
Current sensor U6 is in series on the panel + wire:
PV_POS -> IN+ ... IN- -> PV_TO_CHARGER -> SOLAR IN +.
The solar side shares no conductor with the UNO Q or Grove GND.
Same-name net labels are electrically connected. Each drawn line is one conductor.
Connector pin numbers follow Grove convention, not a left-to-right connector view.
Disconnected: Modulino Movement, Media Carrier, Waveshare USB OUT (5 V / 1 A).
Rejected: HRS1358 + LM2596 step-down feeding the UNO Q (a buck cannot make 5 V from a 3.7 V LiPo).
Wi-Fi phone camera integration remains outside this wiring.
Module / connector view; no PCB footprints
USB-powered UNO Q; separate solar charging path (PR #49)
HackTurin 2026
Sheet: /
File: poopvolt.kicad_sch
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